SQL Assessment Test Answers

Q1 part A: What is total revenue by user acquisition channel?

```sql
SELECT u.channel,
ROUND(SUM(o.revenue), 2) AS total_revenue
FROM orders o
inner JOIN users u
on o.user_id = u.user_id
group by  u.channel
order by total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wr

channel	total_revenue
Paid Ads	163451.68
Referral	132014
Online	130754.64
Organic	84877.36

- Paid Ads generated the highest total revenue among all acquisition channels, indicating strong customer conversion and spending from paid marketing campaigns.
- Organic channel contributed the lowest revenue, suggesting lower customer acquisition or lower spending behavior from organically acquired users.
- Referral and Online channels also showed strong revenue contribution
- The results suggest that Paid Ads is currently the most effective acquisition channel in terms of revenue generation.

Q1 part B: Which channel has the highest average revenue per user?

```sql
SELECT u.channel,
ROUND(SUM(o.revenue) / COUNT(DISTINCT o.user_id),2) AS avg_revenue_per_user
from orders o
join users u
on o.user_id = u.user_id
group by u.channel
order by avg_revenue_per_user DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

channel	avg_revenue_per_user
Online	4086.08
Referral	3474.05
Paid Ads	3335.75
Organic	3264.51

- Online channel has the highest average revenue per user, which shows that users from this channel are spending more on average compared to other channels.

- Even though Paid Ads generated the highest total revenue overall, the spending per user is higher in the Online channel.
- Organic channel has the lowest average revenue per user, which may indicate lower customer spending behaviour from organically acquired users.

Q2 part A: Which countries contributed most to the drop?

```sql
select country,
ROUND(SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END), 2) AS revenue_2024,
ROUND( SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END), 2) AS revenue_2023,
ROUND( SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END) -
       SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END),2) AS revenue_change
FROM orders
group by  country
order by  revenue_change ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	revenue_2024	revenue_2023	revenue_change
USA	43070.55	67589.92	-24519.37
Australia	45770.24	65135.15	-19364.91
India	48739.95	41754.36	6985.59
UK	50113.79	40539.4	9574.39
Germany	67455.27	40929.05	26526.22

- USA contributed the most to the revenue drop from 2023 to 2024.
- Australia also showed a significant decline in revenue.
- India, UK, and Germany showed revenue growth instead of decline.

Q2 part B: Which products are driving the decline?

```sql
select product,
ROUND( SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END), 2) AS revenue_2024,
ROUND( SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END), 2) AS revenue_2023,
ROUND( SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END) -
       SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END),2) AS revenue_change
FROM orders
group by  product
order by revenue_change ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product	revenue_2024	revenue_2023	revenue_change
Laptop	62214.02	71245.47	-9031.45
Phone	53283.98	52918.84	365.14
Tablet	72903.41	70304.47	2598.94
Accessories	66748.39	61479.1	5269.29

- Laptop is the main product driving the revenue decline, with revenue decreasing by around 9K from 2023 to 2024.
- Phone revenue remained almost stable with only a small increase.
- Tablet and Accessories showed revenue growth in 2024 instead of decline.

Q2 part C: Explain your approach before writing SQL

Ans: I compared the revenue for 2023 and 2024 by grouping the data based on country and product. Then I calculated the revenue difference between both years to identify which countries and products contributed most to the decline.

Q3-part A: Write an SQL code to calculate number of users, order value and purchase frequency in 2023 and 2024.

```
SELECT YEAR(STR_TO_DATE(o.order_date, '%d-%m-%Y')) AS year,
COUNT(DISTINCT o.user_id) AS number_of_users,
ROUND( SUM(o.revenue) / COUNT(o.order_id),2) AS avg_order_value,
ROUND(COUNT(o.order_id) / COUNT(DISTINCT o.user_id),2) AS purchase_frequency,
ROUND(SUM(o.revenue), 2) AS total_revenue
FROM orders o
JOIN users u
ON o.user_id = u.user_id
WHERE u.channel = 'Online'
```


WHERE u.channel = 'Online'


```
AND YEAR(STR_TO_DATE(o.order_date, '%d-%m-%Y')) IN (2023, 2024)
GROUP BY YEAR(STR_TO_DATE(o.order_date, '%d-%m-%Y'))
ORDER BY year;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	number_of_users	avg_order_value	purchase_frequency	total_revenue
2023	26	1023.38	2.54	67543.17
2024	26	1036.25	2.35	63211.47

- Total revenue in the Online channel decreased from 2023 to 2024.
- The number of users remained the same, and average order value slightly increased.
- The main reason for the revenue drop is the decrease in purchase frequency, which means users placed fewer orders in 2024.

Q3-part B: Explain your approach before writing SQL.

I filtered the data for the Online channel and compared the performance between 2023 and 2024. To understand the reason behind the revenue drop, I calculated the number of unique users, average order value, and purchase frequency for both years.

Q4

Approach

I compared product revenue between 2023 and 2024 and calculated the percentage change. Based on the change, products were classified as Growing, Stable, or Declining.

```sql
SELECT product,
ROUND(SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END), 2) AS revenue_2023,
ROUND(SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END), 2) AS revenue_2024,
ROUND((SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END) -
    SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END)) /
    SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END) * 100,2) AS growth_percentage,
CASE WHEN ((SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END) -
            SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END)) /
            SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END) * 100) > 10 THEN 'Growing'
WHEN ( (SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END) -
```

```sql
WHEN ( (SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2024 THEN revenue ELSE 0 END) -
        SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END)) /
        SUM(CASE WHEN YEAR(STR_TO_DATE(order_date, '%d-%m-%Y')) = 2023 THEN revenue ELSE 0 END) * 100) BETWEEN -10 AND 10
        THEN 'Stable'
ELSE 'Declining' END AS product_status
FROM orders
GROUP BY product;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product	revenue_2023	revenue_2024	growth_percentage	product_status
Phone	52918.84	53283.98	0.69	Stable
Laptop	71245.47	62214.02	-12.68	Declining
Tablet	70304.47	72903.41	3.7	Stable
Accessories	61479.1	66748.39	8.57	Stable

Q3-part A: **Growing** → revenue increased by more than 10%
No product showed growth above 10%, so none were classified as Growing.

Q4-part B: **Stable** → change between -10% and +10%
Phone, Tablet, and Accessories remained stable as their revenue change stayed within the ±10% range.

Q4-part B: **Declining** → revenue decreased by more than 10%
Laptop is the only declining product, with revenue decreasing by more than 10% in 2024.